Homework #5 — Logistic Regression and Probability; Mid-Semester Survey (15 pts)

MATH 240 — Introduction to Statistics, Fall 2026

Due Date: Friday, 10/16 at 11:59pm ET

Instructions

- Please submit this assignment to Gradescope as two separate .pdf files.
  - Answers to all textbook problems as a .pdf file in the HW5: Textbook Problems dropbox on Gradescope.
  - Knitted Quarto document as a .pdf to the Smoking problem in the HW5: Smoking dropbox on Gradescope.
- Responses can be hand-written or typed for the textbook problems. You will write code with RStudio for Part II.
- Certain questions will ask you to do the problem in a specific way. Please adhere to these instructions or points will be deducted.
- For all questions involving manual calculations, please show all work and round answers to four decimal places.
- For all questions involving code, please provide your code along with the output.
- Refer to syllabus for late work deductions.

Background

We will look at using logistic regression in R and by-hand, as well as exploring some probability concepts. You will also have the opportunity to provide feedback about the course anonymously.

Part I: Textbook Problems (10 pts)

Complete the following problems:

1. Exercises from Chapter 9: #4, #8
2. (3.5) If you flip a fair coin 10 times, what is the probability of
   (a) getting all tails?
   (b) getting all heads?
   (c) getting at least one tails?
3. (3.8) The American Community Survey is an ongoing survey that provides data every year to give communities the current information they need to plan investments and services. The 2010 American Community Survey estimates that 14.6% of Americans live below the poverty line, 20.7% speak a language other than English (foreign language) at home, and 4.2% fall into both categories.
   (a) Are living below the poverty line and speaking a foreign language at home disjoint?
   (b) Draw a Venn diagram summarizing the variables and their associated probabilities.
   (c) What percent of Americans live below the poverty line and only speak English at home?
   (d) What percent of Americans live below the poverty line or speak a foreign language at home?
   (e) What percent of Americans live above the poverty line and only speak English at home?
   (f) Is the event that someone lives below the poverty line independent of the event that the person speaks a foreign language at home?

Part II: Smoking (5 pts)

We will focus on the `smoking` dataset, which has 1,691 observations with 12 variables and contains survey data on smoking habits from the UK.

The dataset exists in the `openintro` package. As such, use these lines of code to load the dataset into R:

- `library(openintro)`
- `data("smoking")`

For this question, you will turn in a separate, knitted Quarto document with code and output related to this question. Create a new Quarto document. Title the document "HW5, Part II" and add your name in the Name box. Note that other packages we have used in the past may and can be used here.

Note: All variables are in their proper formats in R, so the `factor()` function is not needed here. However, the `levels()` function will be useful to identify what is the baseline level (e.g. which category is 0) for a categorical variable.

Answer the following questions below using R code:

(a) Fit a logistic regression model using age, gender, and marital status to predict smoker status. Type out the model with the β coefficients gathered from the fit.

(b) Find the odds ratio for the β coefficient of age. Then, discuss the change in the odds of being a smoker between two people who are a year different in age. Do you have a greater chance of being a smoker if you are older?

(c) Create a stacked bar plot showing the proportions of smokers across different marital statuses. Discuss how these results compare to the results in your fit from part (a).

(d) Find the probability of being a smoker for the following three people:

- A widowed 60-year old male
- A single 25-year old female
- A married 40-year old female

Then, discuss how you would categorize them. Are they smokers or non-smokers?

Optional: Part III: Mid-Semester Survey

Can you believe it? We're halfway through the semester! As we head into fall break, I value feedback from you all regarding what has worked for you in this course and what has not.

Fill out the survey here: https://forms.gle/SxmtGDe6pUXnAZnSA

A Message

No homework will be assigned over fall break! Enjoy the break! :)